

AX-7200 Hydroacoustic Profiler

Operator and Service Manual — firmware 3.4.1. Aldergrove Instruments Ltd., Richmond, British Columbia. Document reference AX-7200-OSM-341.

1. Overview

The AX-7200 Hydroacoustic Profiler is a self-contained hydroacoustic profiling instrument released in March 2021. Product status: Legacy — supported until December 2027.

The instrument records backscatter across the water column and stores profiles internally for later retrieval. It is rated to a maximum operating depth of 600 metres and is intended for deployment from a vessel, a fixed mooring, or an autonomous platform.

This manual covers the AX-7200 only. Procedures for the AX-7300 are similar in outline but differ in several specifics, and the two manuals must not be used interchangeably.

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2. Safety

Read this section before handling the instrument. The pressure housing stores significant energy when pressurised. Never attempt to open the housing while it is pressurised or immediately after recovery from depth.

- Vent the housing using the bleed screw before opening.

- Do not exceed the rated operating depth, including transient excursions during recovery.
- The battery pack contains lithium-ion cells. Do not puncture, crush, or expose to temperatures above the rated maximum.
- Disconnect power before servicing the connector.

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3. Specifications

Model	AX-7200
Maximum operating depth	600 m
Operating temperature range	-5 °C to 45 °C
Battery endurance	18 hours continuous
Charge time	4.5 hours
Weight in air	4.2 kg
Sampling rate	200 kHz
Beam width	6.5 degrees
Internal storage	64 GB

Housing material	anodised 6061-T6 aluminium
Connector	SubConn MCBH8M
External power input	12–24 V DC
Firmware version	3.4.1
Warranty period	24 months
Calibration interval	12 months

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4. Installation

4.1 Mechanical mounting

Mount the instrument with the transducer face unobstructed and at least 300 mm from any reflective surface. Use the supplied clamp set; do not clamp directly onto the transducer housing.

4.2 Electrical connection

The instrument accepts 12–24 V DC through the SubConn MCBH8M connector. Apply a thin film of the supplied silicone grease to the connector pins before mating. Mate the connector fully until the locking sleeve seats.

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5. Operation

5.1 Pre-deployment checks

- Confirm firmware is at version 3.4.1 or later.
- Confirm the battery indicates a full charge; a full charge takes approximately 4.5 hours.
- Confirm the internal clock is synchronised to UTC.
- Confirm available storage is sufficient for the planned deployment duration.
- Inspect the O-ring seat for debris and replace the O-ring if in any doubt.

5.2 Deployment

At the configured sampling rate of 200 kHz, a fully charged instrument records continuously for approximately 18 hours. Reducing the ping rate extends endurance approximately linearly.

5.3 Recovery and data retrieval

Rinse the instrument in fresh water on recovery and allow it to dry before opening. Retrieve data over the wired interface; wireless retrieval is not supported for full-resolution profiles.

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6. Maintenance and calibration

Return the instrument for factory calibration every 12 months. Calibration outside this interval invalidates the accuracy specification and may invalidate warranty claims relating to measurement performance.

Inspect and replace the main O-ring at every deployment cycle. Inspect the anode annually and replace it when 50 per cent consumed. Clean biofouling from the transducer face with a soft brush and fresh water only — do not use solvents or abrasives.

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7. Troubleshooting

The instrument reports faults as numbered error codes on the status display and in the deployment log file.

E-119	Transducer calibration drift beyond tolerance. Re-run the bench calibration routine and, if the fault persists, return the head for factory recalibration.
E-204	Battery thermal cutoff. The pack has exceeded 52 °C and has disconnected itself. Allow 30 minutes of cooling before restarting.
E-142	Pressure sensor reading outside the expected envelope for the reported depth. Inspect the port for biofouling.

If a fault persists after the action described above, record the code, the firmware version, and the deployment log, and contact technical support before redeploying the instrument.

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8. Warranty

The AX-7200 is supplied with a 24-month warranty from the date of delivery, covering defects in materials and workmanship. The warranty excludes consumables — O-rings, anodes, and the battery pack after twelve months — and excludes damage caused by operation outside the rated depth or temperature envelope.

Warranty claims require the instrument serial number and the deployment log covering the failure.

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9. Appendix — part numbers

AX7200-BC-SPARE-ORING	Main housing O-ring, pack of five
AX7200-BC-SPARE-ANODE	Sacrificial anode
AX7200-BC-BATT	Replacement battery pack
AX7200-BC-CLAMP	Mounting clamp set
AX7200-BC-CABLE-5M	Deck cable, 5 metre

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